

Course Content Outline and Subtopics		CLO*	Learning and Teaching Activities**								Total SLT	
			Face-to-Face (F2F)									
			Physical				Online/ Technology-mediated (Synchronous)					NF2F Independent Learning (Asynchronous)
L	T	P	O	L	T	P	O					
1	Fundamentals of Computer Architecture • Introduction to computer systems • Basic components (ALU, CU, memory, I/O) • Von Neumann vs Harvard architecture • Stored program concept • Classification of computers	1	4	1							5	10
2	Number Systems and Digital Codes • Binary, Octal, Decimal, Hexadecimal number systems and interconversions • Gray Code • BCD (Binary Coded Decimal) • Application of digital codes	2	3	1							4	8
3	Digital logic and circuit design • Logic gates • Combinational & sequential circuits • Half adder, Full Adder • K-Map simplification (3 and 4 variable)	2	4	1							5	10
4	Processor Organization and Instruction cycle • CPU internal structure • ALU, Control Unit, Register • Types of registers • Instruction execution cycle, machine cycle • System Bus (data, address, control buses)	3	3	2							5	10

5	Memory Architecture and Hierarchy • Memory hierarchy • RAM & ROM types • Address decoding • Cache Memory: locality of reference, hit/miss ratio, mapping techniques • Associative memory concepts	3	4	2							6	12
6	Instruction Set and Addressing Modes • Instruction formats • Addressing modes • Data movement and arithmetic instructions • RISC vs CISC • Stack/Accumulator/Register-Memory architectures	3	3	2							5	10
7	Pipelining and Parallelism • Basics of pipelining • Pipeline types • Pipeline hazards • Parallel computing concept • Flynn’s classification • Instruction-level parallelism • Performance benefits and limitations	4	4	1							4	9
8	Input/Output Systems • I/O device interfacing • I/O-mapped vs memory-mapped I/O • Interrupt handling • DMA (Direct Memory Access) • Handshaking and communication control	4	3	2							3	8
9	Modern Storage Systems and Trends • Overview of SSDs and HDDs • Magnetic and optical media • RAID, persistent memory, cloud-based storage • Storage speed comparison • NVMe and flash trends	5	3	2							3	8
SUB-TOTAL SLT:												85

